

Patrick D. VanMeter, PhD

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PROFESSIONAL SUMMARY

Versatile scientist with cross-disciplinary interests and substantial experience using computational modeling, statistics, physical principles, and imaging tools to analyze data and solve complex problems in physics and healthcare.

Diverse set of technical, analytical, and interpersonal skills applied with rigorous scientific mindset and intuition, resulting in numerous peer-reviewed scientific publications and professional accomplishments.

Effective at working in a multidisciplinary team environment; participated in international collaborations with labs in the US, Italy, France, and Germany, while meeting deadlines and achieving personal milestones.

EDUCATION

PhD in Physics

University of Wisconsin–Madison, 2020

Advisors: D. Den Hartog & J. Sarff

MA in Physics

University of Wisconsin–Madison, 2017

Emphasis in Plasma Physics

BS in Mechanical Engineering & Physics

University of Kentucky, 2014

Minor in Mathematics, *summa cum laude*

EXPERIENCE

Developing innovative applications of machine learning to ultrasound imaging

as a Sr. Data Scientist at GE Healthcare (Remote, 2023 – Present)

- Train and evaluate deep learning computer vision models for segmentation of lung ultrasound images.
- Created a sophisticated post-processing pipeline for processing and interpreting segmentation model outputs.
- Transferred and organized large sets of medical image data from clinical sites, and worked with doctors and sonographers to create and review ground-truth labels.
- Served as the primary technical team member while conducting verification and validation tests for an FDA submission.

Machine learning model research and development for healthcare analytics

as a Data Scientist at Certilytics (Remote, 2021 – 2023)

- Explored new methods for evaluating data representations in healthcare predictive models.
- Created a procedural grouper pipeline for claims using interconnected machine learning models.
- Designed, developed, and deployed deep learning models for healthcare analytics by adapting techniques from natural language processing.

Improving clinical outcomes with advanced medical imaging and AI technologies

as a Postdoctoral Research Fellow at the Mayo Clinic CT Clinical Innovation Center (Rochester, MN, 2020 – 2021)

- Work with a multidisciplinary team to develop novel applications of photon-counting computed tomography and deep learning to cardiac and renal imaging, with a focus on improving clinical outcomes.
- Develop an image de-noising/de-convolution pipeline based on a convolutional neural network architecture.
- Designed experiments, operated CT scanners, performed reconstructions, and analyzed the resulting images.
- Presented at 3 scientific conferences and contributed to multiple scientific papers.

Hardware development, statistical modeling, and integrated data analysis

as a Graduate Research Assistant at the Wisconsin Plasma Physics Lab, UW–Madison (Madison, WI, 2015 – 2020)

- Developed an x-ray diagnostic forward model in Python for use in a Bayesian integrated data analysis framework and applied this to a large data set to analyze the dynamic behavior of fusion plasmas.
- Developed Python software to calibrate the detector, prepare it for operation, take data, and write to the database.
- Results presented at 7 conferences and published in 5 first-authored and 5 co-authored scientific papers.

Physics classroom and laboratory instruction

as a Teaching Assistant at the University of Wisconsin–Madison (Madison, WI, 2014 – 2015)

- Instructed students, created and graded assignments, and facilitated labs for introductory Physics I and II courses.

PUBLICATIONS

J. Marsh, **P. D. VanMeter**, K. Rajendran, S. Leng, and C. McCollough. “Ex vivo coronary calcium volume quantification using a high-spatial-resolution clinical photon-counting-detector CT,” *Journal of Medical Imaging* Vol. 10, Issue 4, 043501 (2023).

L. F. Delgado-Aparicio, D. Del-Castillo-Negrete, N. C. Hurst, **P. D. VanMeter**, *et al.* “Off-axis runaway-electron seed formation, growth and suppression,” *arXiv:2204.05353* (2022).

L. F. Delgado-Aparicio, **P. D. VanMeter**, T. Barbui, O. Chellai, *et al.* “Multi-energy reconstructions, central electron temperature measurements, and early detection of the birth and growth of runaway electrons using a versatile soft x-ray pinhole camera at MST,” *Review of Scientific Instruments* 92 073502 (2021).

P. D. VanMeter, L. F. Delgado-Aparicio, L. M. Reusch, and D. J. Den Hartog. “Robust analysis of space-, time-, and energy-resolved soft x-ray measurements of magnetically confined fusion plasmas,” *Review of Scientific Instruments* 92, 053549 (2021).

P. D. VanMeter, “Observations of helical plasma dynamics using complementary x-ray diagnostics in the MST,” PhD Thesis, *University of Wisconsin–Madison* (2020).

P. D. VanMeter, L. F. Delgado-Aparicio, L. M. Reusch, and D. J. Den Hartog. “A versatile multi-energy soft x-ray diagnostic for T_e measurements in the Madison Symmetric Torus,” *Journal of Instrumentation* 14 C09009 (2019).

L. Reusch, P. Franz, D. J. Den Hartog, J. A. Goetz, M. D. Nornberg, and **P. VanMeter**. “Model Validation for Quantitative X-Ray Measurements,” *Fusion Science and Technology* 74 167 (2018).

P. VanMeter, L. F. Delgado-Aparicio, J. Wallace, L. Reusch, *et al.* “Pixel-to-pixel variation on a calibrated PILATUS3-based multi-energy soft x-ray detector,” *Review of Scientific Instruments* 89 10G119 (2018).

L.F. Delgado-Aparicio, J. Wallace, H. Yamazaki, **P. VanMeter**, *et al.* “Simulation, design, and first test of a multi-energy soft x-ray (SXR) pinhole camera in the Madison Symmetric Torus,” *Review of Scientific Instruments* 89 10G116 (2018).

P. D. VanMeter, P. Franz, L. M. Reusch, J. S. Sarff, and D. J. Den Hartog. “A comparison between soft x-ray and magnetic phase data on the Madison Symmetric Torus,” *Review of Scientific Instruments* 87 11E322 (2016).

PRESENTATIONS

“Quantification of coronary calcification using high-resolution photon-counting-detector CT and an image domain denoising algorithm.” *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference*, San Diego, CA, 2022.

“Determination of *in-vivo* kidney stone mixed composition using high-resolution photon counting CT with AI-based denoising and deconvolution.” *Radiological Society of North America (RSNA) Annual Meeting*, Chicago, IL, 2021.

“Evaluation of Image-Domain Training Frameworks for Deep Learning Based Denoising and Deconvolution.” *American Association of Physicists in Medicine (AAPM) Annual Meeting*, Remote, 2021.

“Robust analysis of space-, time-, and energy-resolved soft x-ray measurements of magnetically confined fusion plasmas.” Invited talk. *High Temperature Plasma Diagnostics (HTPD) Conference*, Remote, 2021.

TECHNICAL SKILLS

- Programming languages: Python, C++, MATLAB, Java, R, SQL.
- Machine learning software: Jupyter, scikit-learn, TensorFlow/Keras, PyTorch.
- Data labeling platforms: Labelbox and V7 Darwin
- Software tracking and version control, including Git and MLflow.
- Statistics, deep learning, natural language processing, computer vision, and data visualization.
- Topics in computer vision, including image processing, segmentation, and convolutional neural networks.
- Physics and applied mathematics at the graduate level.
- Familiarity with X-ray computed tomography (CT) and ultrasound imaging.
- Experimental design & analysis.